

		$N = mg + \frac{mv^2}{r} = mg + 2mg$ $= 3mg$
10	Taking the Earth to be a perfect sphere of uniform density rotating about its polar axis, which of the following statements concerning the observed acceleration due to free fall, a , at the surface of the Earth is true?	
	A	The value of a at the equator is larger than that at the poles.
	B	If the rate of rotation of the Earth slows down, a at the equator increases.
	C	If the radius of the Earth increases with its density remaining unchanged, a at the poles decreases.
	D	If the radius of the Earth increases with its density remaining unchanged, a at the equator decreases.

	Ans: B $a = g - r\omega^2$
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